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Predicting Red Wine Preferences Based on Physicochemical Properties

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Introduction

- Wine classification is a difficult task because wine savouring requires a combination of taste, smell, texture and so on
- Challenge wine producers to determine which type of wine should be focused on to target their consumers.
- This report investigates the linear relationship between consumers' red wine preferences and the different physicochemical properties.
- It will result in a multiple linear regression model that wine producers can use to predict the trend in consumers' preferences and improve their wine-making and selling process.
- Give me novel knowledge about wine and what type of red wine is on trend in the market.

Research Question

What is the contribution of different wine's physicochemical properties in the consumers' red wine preference?





Background and Literature

- Google Scholar was used to search for background information and peer-reviewed papers related to wine.
- Keywords: red wine, wine preference, wine industry, sensory analysis of wine
- Approximately 2 million results come up for the first 3 keywords, and over 300,000 results for the last one
- Overall, decision on wine is based on only a small amount of available information. While taste is the dominating factor, quality is difficult to define and communicate because of its subjectivity

Background and Literature

1, Rethinking The Wine List: Restaurant Customers' Preference for Listing Wines According To Wine Style (Staub & Siegrist, 2022)

- Examined whether constructing wine list according to taste or origin would facilitate customers' decision-making on wine.
- Participants were given style-grouped or origin-grouped wine lists to assess their choosing style preferences.
- Result: Style-grouped wine list was chosen by more than half of the participants
=> Many consumers may value wine lists that are arranged by the taste of wine.
- Knowledge implementation in my study:
 - Taste is a key indicator when it comes to customers' wine preferences, so it is crucial to go deeper into chemical structures that construct the taste of wine.
 - Need variables focusing on taste-producing.
 - Since it was clear that style matters more than origin, my report will expand on this topic.

Background and Literature

2, Consumer Preferences for Different Red Wine Styles and Repeated Exposure Effects (Tempere et al., 2017)

- Investigated the effect of wine exposure on consumers' appreciation for two contrasting wine styles (fresh/jammy fruit and low/high alcohol concentration).
- One-way and two-way ANOVA test was used.
- Result:
 - Daily exposure changes consumers' appreciation of wine.
 - Fruity expression and ethanol concentration were crucial factors in determining wine preferences.
- Knowledge implementation in my study:
 - There are many other factors that can affect preference for wine besides physicochemical properties.
 - Since fruity level acts as an important indicator, more focus on chemicals that produce fruity flavours should be found (e.g. citric acid).
 - Instead of examining general taste, my study will focus on the combination of physicochemical properties and expand on the taste aspect of wine.

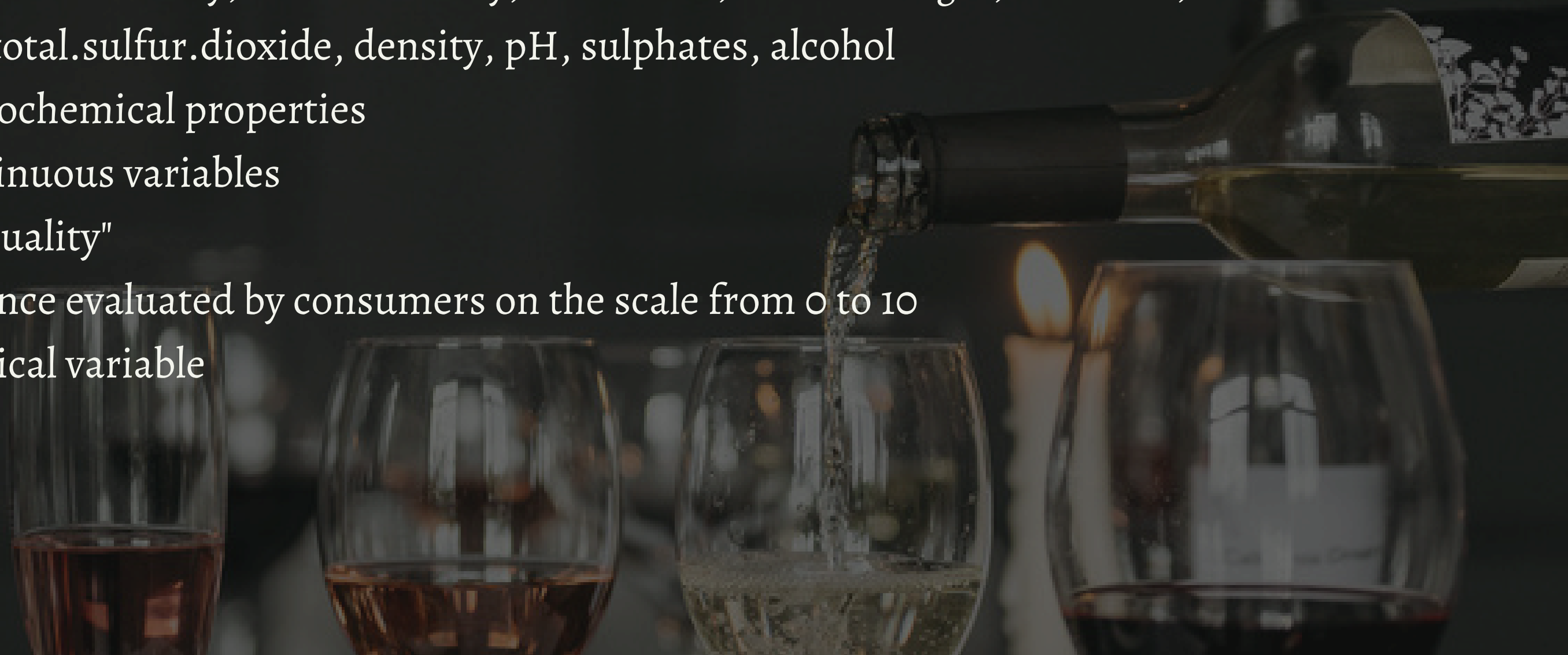
Background and Literature

3, Modeling Wine Preferences by Data Mining from Physicochemical Properties (Cortez et al., 2009)

- Predicted consumers' wine preferences from physicochemical properties information that was available at the wine quality certification step.
- Four fuzzy hybrid techniques were used.
- Result: Alcohol, fixed acidity, free sulfur dioxide, residual sugar and volatile acidity have the highest causal relation with wine quality, respectively.
- Knowledge implementation in my study:
 - This study is closely related to my research question.
 - Gave me an idea of what elements to search for in a data set.
 - My report will rigorously expand on this topic by implementing a quantitative multiple linear regression model.

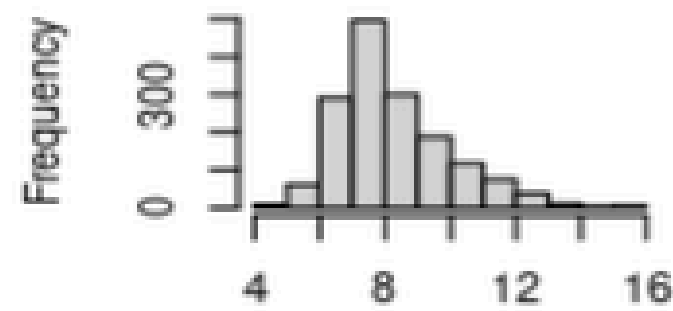
Data Description

- *Wine Quality Data Set* was found on Kaggle, but the original source is from the UCI machine learning repository, which includes red wine variants of the Portuguese "Vinho Verde" wine.
- Data set includes 12 variables with 1599 observations.
 - The first 11 variables: fixed.acidity, volatile.acidity, citric.acid, residual.sugar, chlorides, free.sulfur.dioxide, total.sulfur.dioxide, density, pH, sulphates, alcohol
 - Names of physicochemical properties
 - Sequential, continuous variables
 - The last variable is "quality"
 - Levels of preference evaluated by consumers on the scale from 0 to 10
 - Ordinal, categorical variable

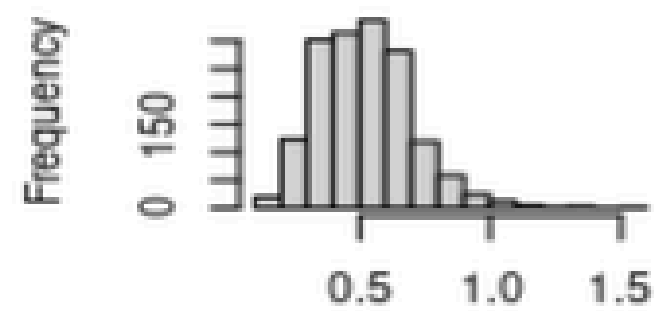


Data Description

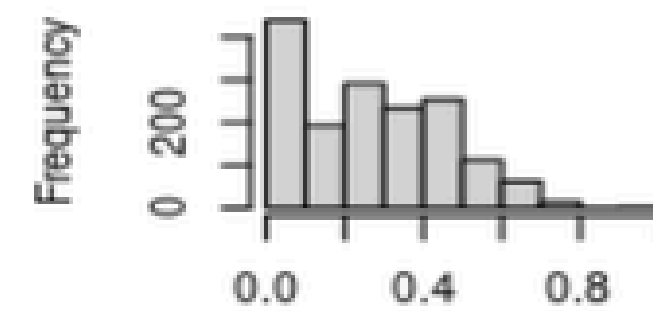
Histograms
showing the
amount of
each
physicochemi-
cal property
in red wine



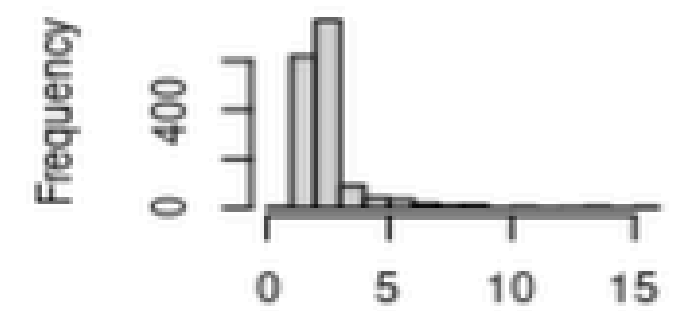
Fixed Acidity



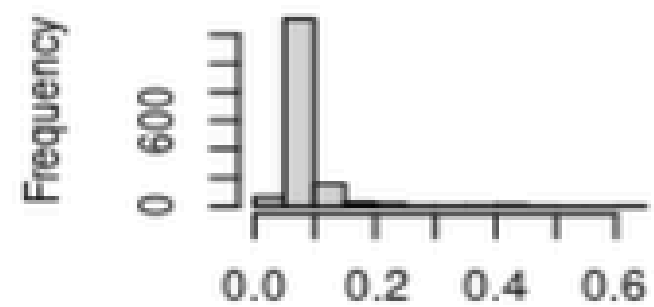
Volatile Acidity



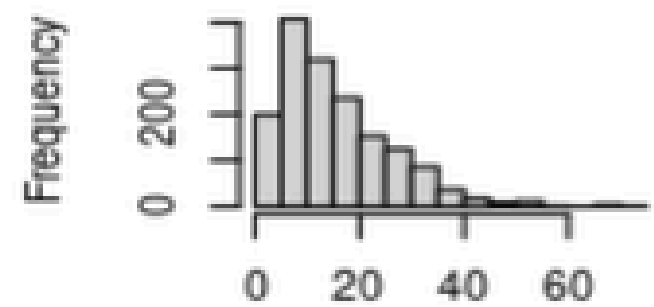
Citric Acid



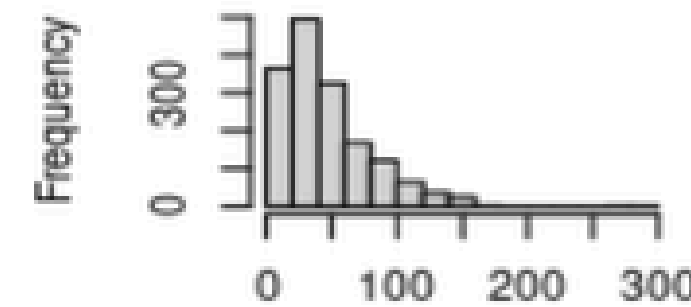
Residual Sugar



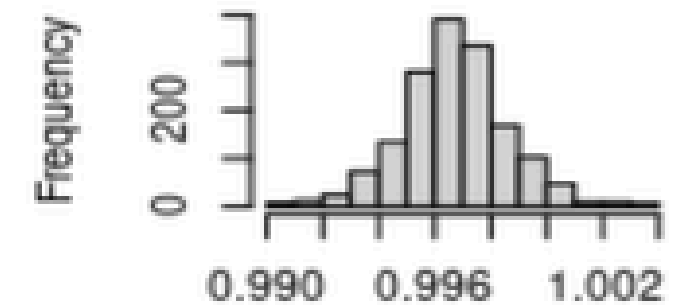
Chlorides



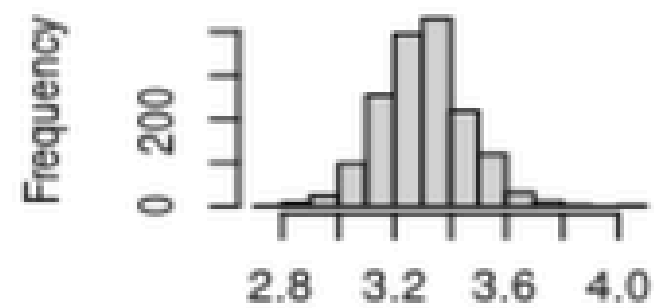
Free Sulfur Dioxide



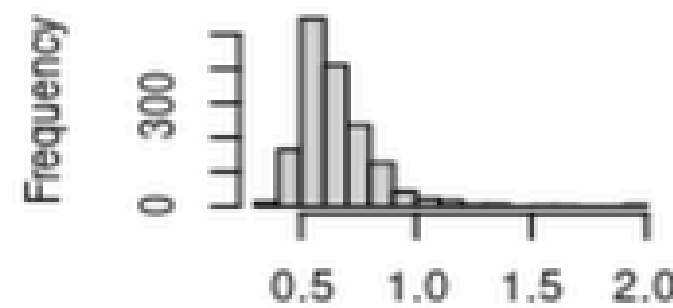
Total Sulfur



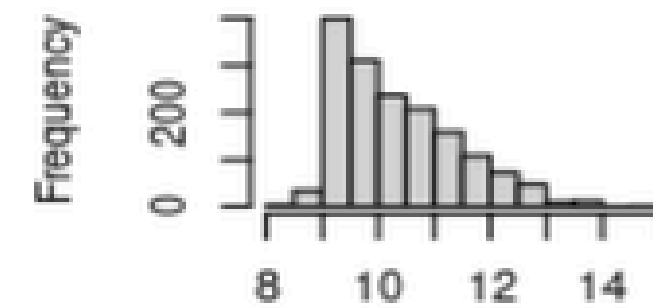
Density



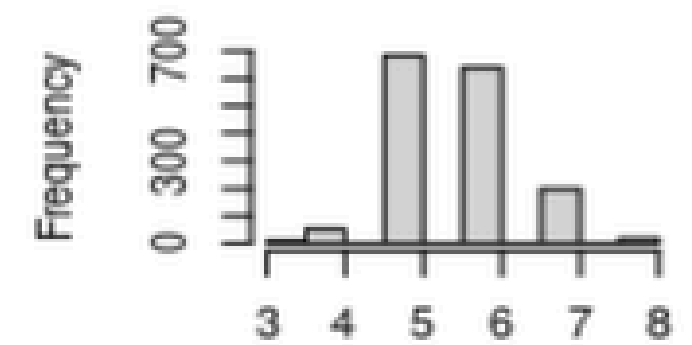
pH



Sulphates



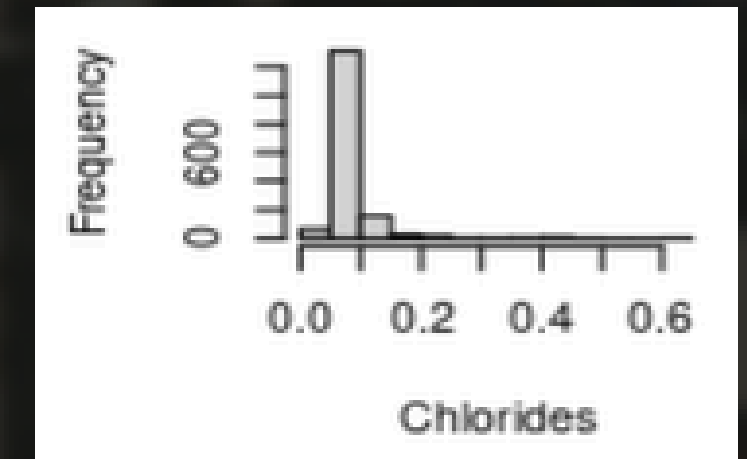
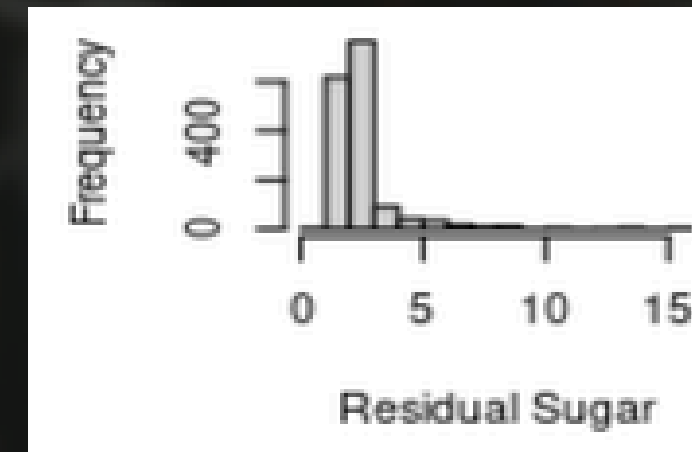
Alcohol



Quality

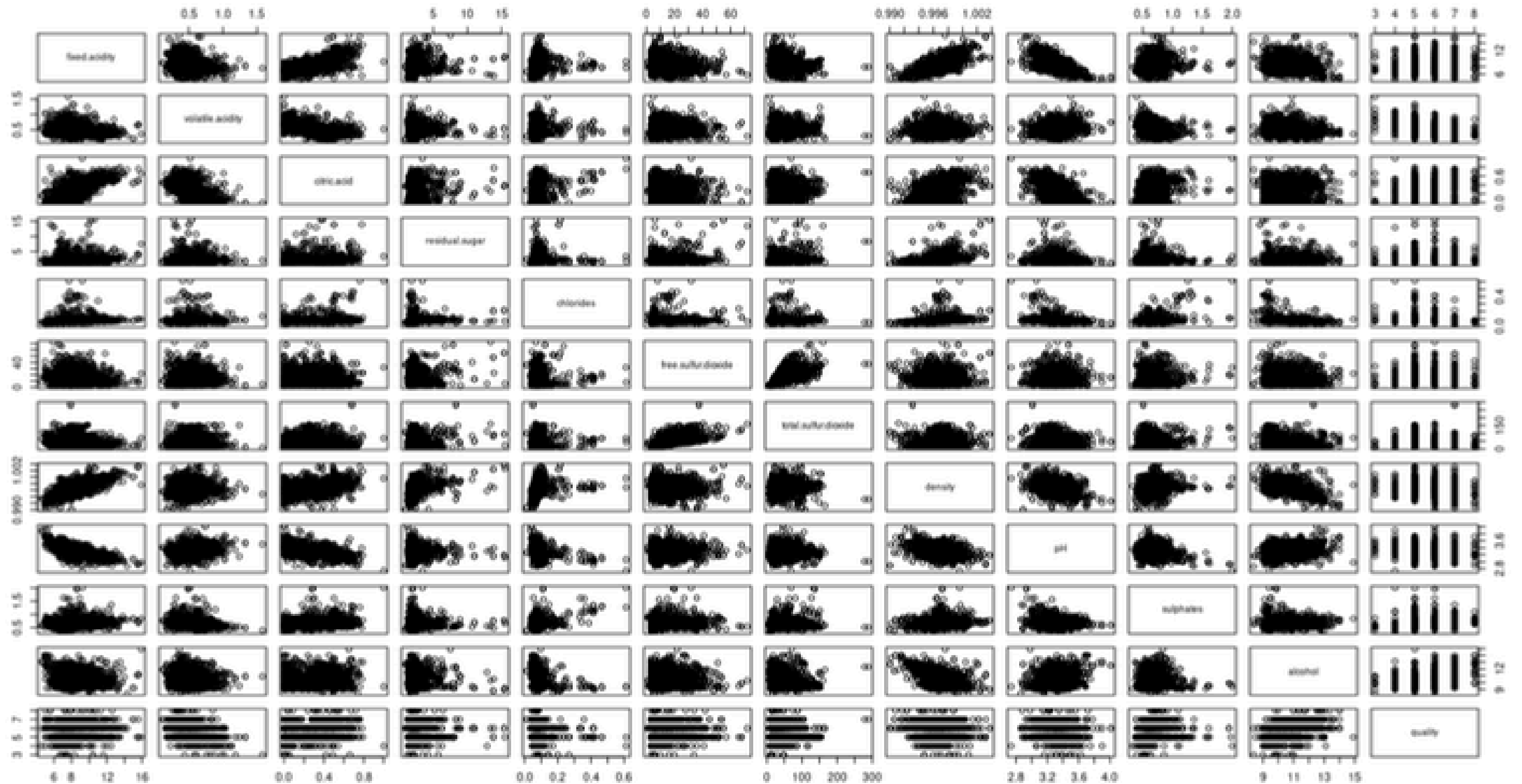
Data Description

- Most of the distributions of physicochemical properties are left-skewed, indicating that the concentration in each element is not very high.
- There exists some large outliers in Residual Sugar and Chlorides.
- On the scale from 0 to 10 in Quality, consumers mostly choose 5 and 6, highlighting that their satisfaction with the wine is somewhat above the average



Data Description

Scatterplots
showing the
relationship
between each
variable in
the Red Wine
Data Set



Data Description

Correlation coefficients between each variable in the Red Wine Data Set

	fixed.acidity	volatile.acidity	citric.acid	residual.sugar	chlorides	free.sulfur.dioxide	total.sulfur.dioxide	density	pH	sulphates	alcohol	quality
fixed.acidity	1.000	-0.256	0.672	0.115	0.094	-0.154	-0.113	0.668	-0.683	0.183	-0.062	0.124
volatile.acidity	-0.256	1.000	-0.552	0.002	0.061	-0.011	0.076	0.022	0.235	-0.261	-0.202	-0.391
citric.acid	0.672	-0.552	1.000	0.144	0.204	-0.061	0.036	0.365	-0.542	0.313	0.110	0.226
residual.sugar	0.115	0.002	0.144	1.000	0.056	0.187	0.203	0.355	-0.086	0.006	0.042	0.014
chlorides	0.094	0.061	0.204	0.056	1.000	0.006	0.047	0.201	-0.265	0.371	-0.221	-0.129
free.sulfur.dioxide	-0.154	-0.011	-0.061	0.187	0.006	1.000	0.668	-0.022	0.070	0.052	-0.069	-0.051
total.sulfur.dioxide	-0.113	0.076	0.036	0.203	0.047	0.668	1.000	0.071	-0.066	0.043	-0.206	-0.185
density	0.668	0.022	0.365	0.355	0.201	-0.022	0.071	1.000	-0.342	0.149	-0.496	-0.175
pH	-0.683	0.235	-0.542	-0.086	-0.265	0.070	-0.066	-0.342	1.000	-0.197	0.206	-0.058
sulphates	0.183	-0.261	0.313	0.006	0.371	0.052	0.043	0.149	-0.197	1.000	0.094	0.251
alcohol	-0.062	-0.202	0.110	0.042	-0.221	-0.069	-0.206	-0.496	0.206	0.094	1.000	0.476
quality	0.124	-0.391	0.226	0.014	-0.129	-0.051	-0.185	-0.175	-0.058	0.251	0.476	1.000

Data Description

- In general, the relationship between each element is not clear, highlighting the potential to see a normality violation, as well as the potential to see linearity problems or maybe just poorly fitting models
=> As a result, looking at the correlation coefficient model, most of them are close to 0 which indicates that there might not be a linear relationship between many physicochemical properties and consumers' wine preferences
- In the scatterplot, the response ("quality") is clustered in groups of answers due to the fact that it is a categorical variable

Discussion

- Correlation coefficient measures how well a predictor and response form a linear relationship with each other, which can range from -1 (negative linear relationship) to 1 (positive linear relationship).
- Although from EDA, we see that there are a lot of correlation coefficients close to 0, the purpose of the study is to figure out which physicochemical property has the highest influence on consumers' preference => Multiple linear regression model is still an appropriate tool to answer the research question.
- Residual Sugar and Chlorides have extremely skewed distributions, so these two predictors are most likely to violate the constant variance assumption (holding all other predictors the same).

References

- Cortez, P., Cerdeira, A., Almeida, F., Matos, T., and Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. In *Decision Support Systems*, Elsevier, 47(4):547-553. Retrieved October 22, 2022, from <https://www.scitepress.org/Papers/2015/55519/55519.pdf>
- Tempere, S., Pérès, S., Espinoza, A. F., Darriet, P., Giraud-Héraud, E., & Pons, A. (2018, December 17). Consumer preferences for different red wine styles and repeated exposure effects. *Food Quality and Preference*. Retrieved October 22, 2022, from https://www.sciencedirect.com/science/article/pii/S0950329318303021?casa_token=kUc7Ho15cNIAAAAAA%3ARgWLeDBobuWu9cW1Bvhht_Bae6frZheFHJQ5dx2nzU4zx3vDRze4zD7DUdIsLvbnAFaxHn5H2Bo
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